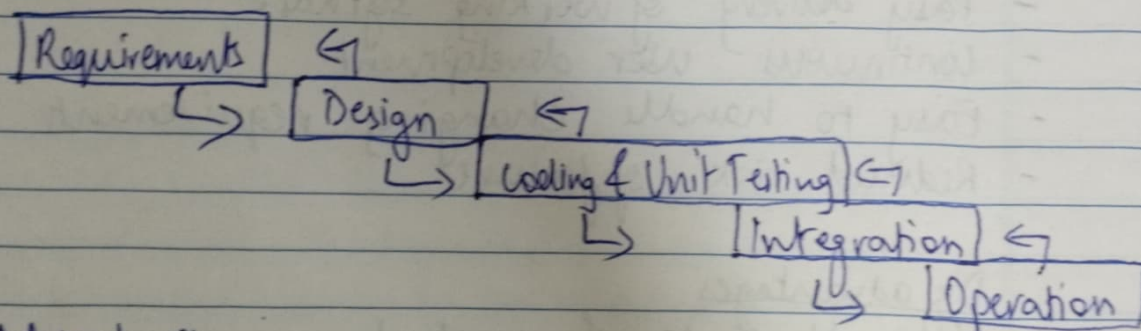


SEPM Assignment 1

1. Software Process Models

1. Waterfall model: It is a sequential process where software development progresses through fixed phases.



Advantages

- Easy to understand & implement.
- Well defined stages with clear milestones
- Proper documentation at every stage
- Suitable for small or well defined projects.

Disadvantages

- difficult to accommodate changes after phase completion
- Testing is done late in the process
- not suitable for complex or long term projects

2. Evolutionary / Exploratory model

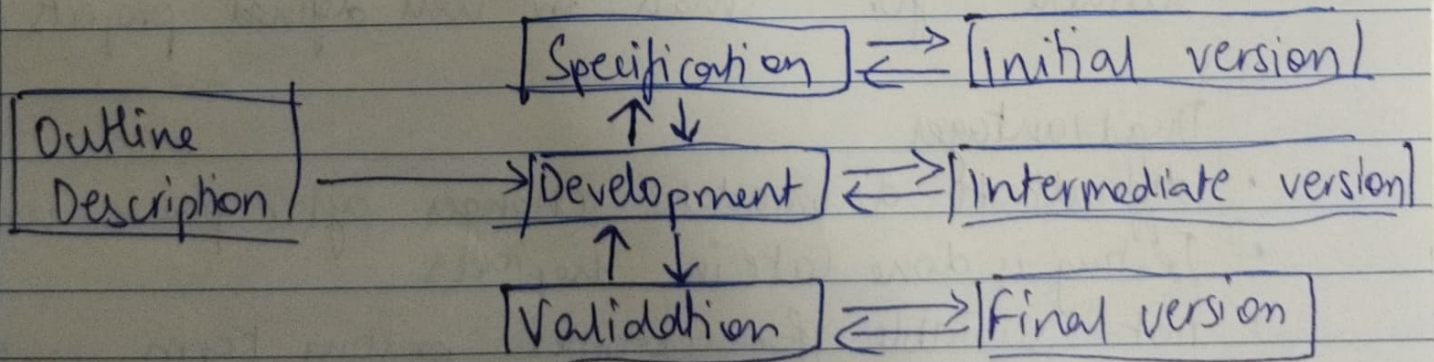
In this model software is developed in multiple versions. The initial version provides base flexibility whereas later versions are enhancements based on user feedback.

Advantages

- Easy delivery of working software
- Continuous user development
- Easy to handle changing requirements
- Reduced risk of failure.

Disadvantages

- Difficult to plan & control
- May result in poor system design
- Documentation may be incomplete.
- Requires experienced developers



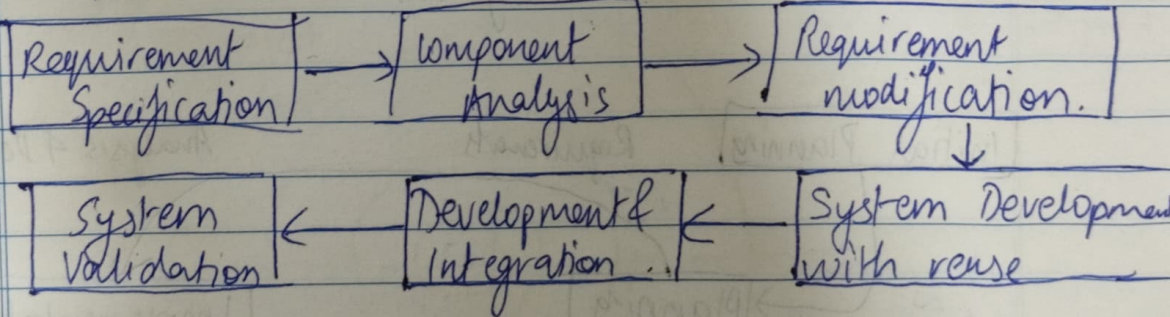
3 Component based Software Engineering focuses on building systems by reusing existing software components.

Advantages

- Faster development time
- Reduced development & maintenance cost
- High reliability due to tested components
- Encourages software reuse.

Disadvantage

- Limited control over reused components
- compatibility & integration issues
- Performance may be affected.



4 Incremental model

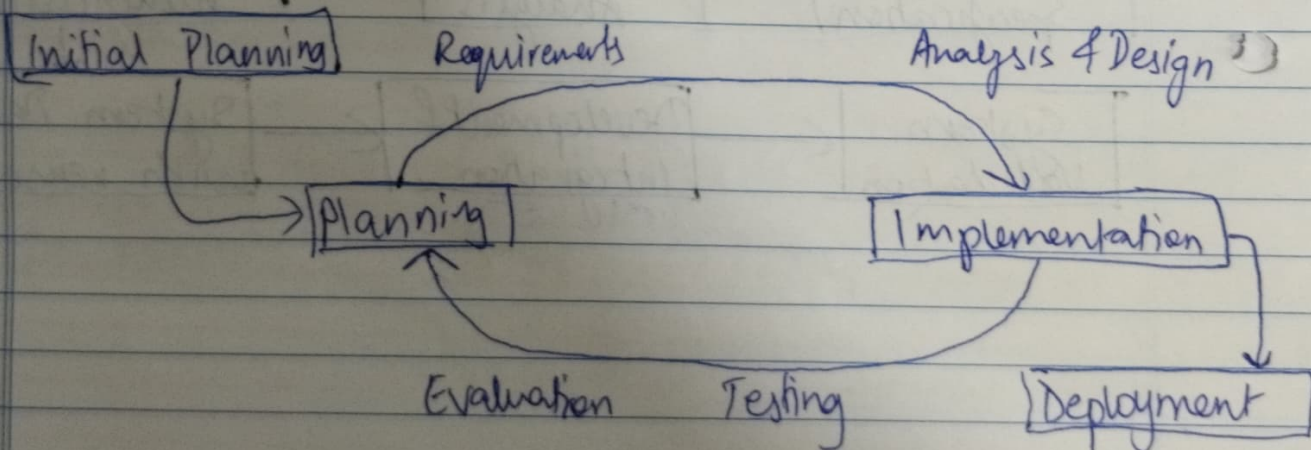
software is developed & delivered in small increments. Each increment adds new functionalities.

Advantages

- Early delivery of partial working software
- Easier testing & debugging
- Flexible to req. changes.
- Reduced risk compared to waterfalls.

Disadvantages

- Requires good planning & design
- Integration of increments can be complex
- Total cost may increase



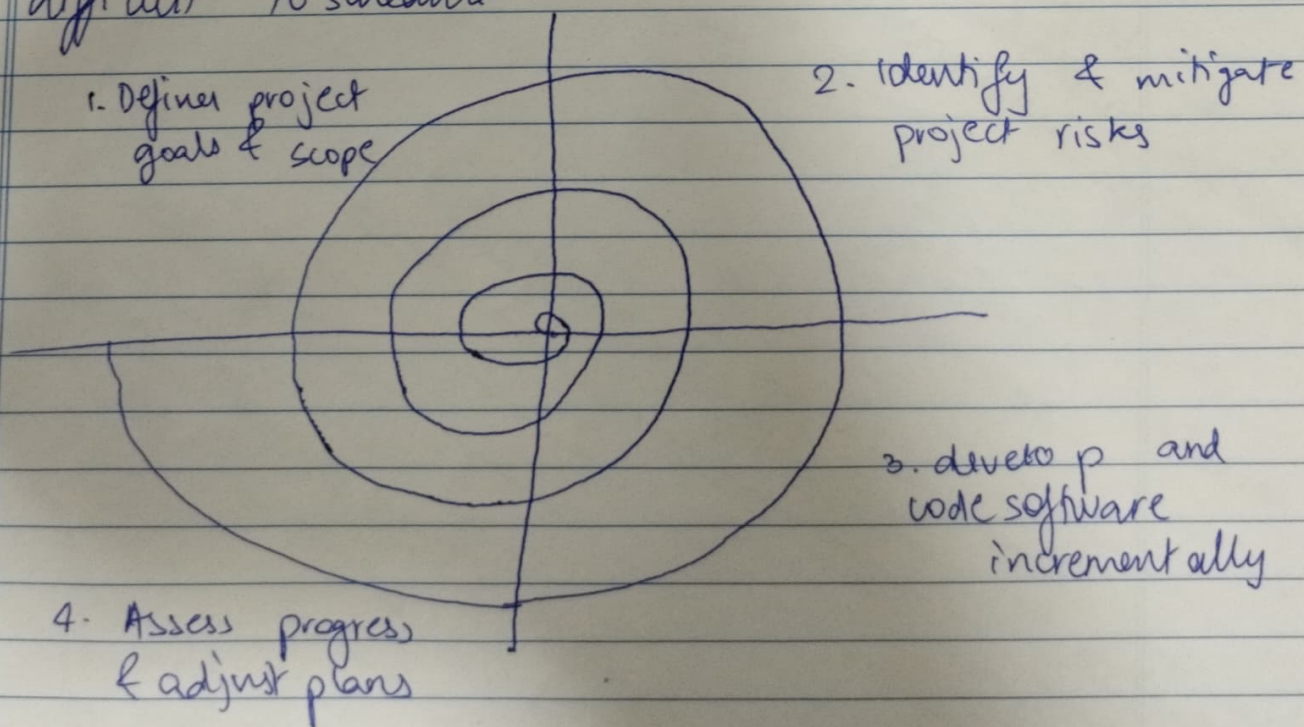
5 Spiral Model is a risk driven process combining iterative development with ~~iterative~~ systematic risk analysis.

Advantages

- Effective risk management
- Suitable for large & complex projects
- continuous user feedback

Disadvantages

- Expensive & complex to manage
- Requires expert risk assessment
- not suitable for small projects
- difficult to schedule.



- 6 Agile model : iterative & incremental approach focused on rapid development, continuous feedback & close collaboration with customers.

Advantages

- Highly flexible
- Improved product quality

Disadvantages

- less documentation
- difficult to predict time & cost

